

AVANTA

Naming the vessel behind an oil slick, from radar, AIS and forward drift simulation — and declining to name one when the evidence does not support it.

CONSOLE

avanta.spacesdrive.cc

API

avantaapi.spacesdrive.cc

READ TIME

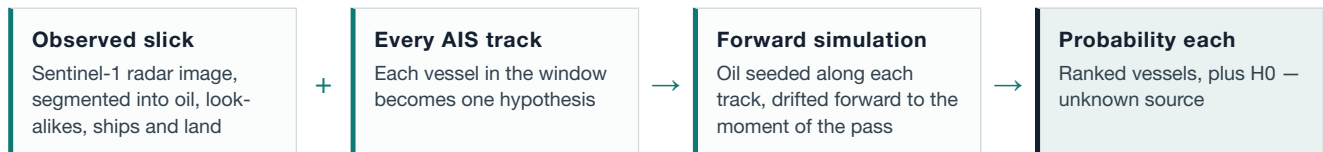
About five minutes

TEAM

Avanta · SIH26143

01 What AVANTA does

Radar sees an oil slick. It cannot see who caused it. AVANTA closes that gap by turning the question round: instead of running the drift backwards from the oil, it takes every ship that was in the area and simulates forward, for each one, the slick that ship *would have* left along its own AIS track. Each simulated slick is compared with the one the satellite actually recorded, and the result is a probability per vessel — alongside an explicit "unknown source" option, so the system can decline to accuse anyone at all.



The drift always runs forwards. Running it backwards cannot work: turbulent diffusion is a random walk and a random walk has no inverse, and the oil you can see has already weathered, so it is not the object that was released.

02 The five-step workflow

The console header carries these five steps as a progress rail, so you can always see where in the chain you are.

1

Watch — pick something to investigate

YOU SEE A full-width map of the area with acquisition footprints, an acquisition ledger of scenes already ingested, and a **Run demo scenario** deck of three one-click cases. Each card carries its own data-mode badge and an honesty label.

YOU DO Click a scenario card, or select a row in the ledger and press Enter.

YOU GET A progress rail naming the current stage, then the console moves itself to the Scene.

2

Detect — read the scene and its reasoning

YOU SEE The calibrated radar image with detected regions drawn over it. On the right, a **Wind gate** card showing the measured wind speed, its direction and the 3–10 m/s band, marked **PASSED** or **GATED**; a swath-coverage card if only part of the box was imaged; a **Detected regions** list; and **Why each region was classified**, giving the five discriminating tests with their values, thresholds and weights.

YOU DO Read the gate verdict first, then the region reasoning. Then press **FIND CANDIDATE VESSELS**.

YOU GET A table of every AIS track scored on geometry — perpendicular distance to the slick's own axis, course alignment, whether drift could physically have carried oil across in the time available. Radar contacts with no AIS at all are kept unconditionally. If the button is greyed out, a sentence beneath it says exactly why.

3

Attribute — watch the physics and read the ranking

YOU SEE A ranked list of candidates with a probability bar each, and the **Ho — unknown source** row sitting in the same ranking. Below the map, a timeline scrubber. The overlay toggle cycles *observed*, *simulated*, *both* and *difference*.

YOU DO Click a candidate row. Press Space to play the particle record, or drag the *Simulation time* slider to step through the seeding phase by hand. Cycle the overlay to *difference* — that is what the likelihood actually scores.

YOU GET Oil entering the water along the track over time, not from a point, then drifting forward under real wind and currents to the acquisition instant.

4

Evidence — check why that number

YOU SEE A drawer of signed contribution bars, one per term.

YOU DO Press `OPEN EVIDENCE BREAKDOWN`, or the E key. Esc closes it.

YOU GET Every likelihood and prior term with its raw value: coverage of the observed oil, the penalty for simulated oil that landed where none was seen, the ensemble adjustment, and each behaviour-prior feature with its weight. They sum to the reported score, so nothing arrives unexplained.

5

Dossier — produce something an officer can send

YOU SEE A MARPOL Annex I, Appendix 3 evidence dossier, laid out field by field against the IMO's itemised list, followed by a provenance block and a manifest hash.

YOU DO Press `GENERATE DOSSIER`, then `DOWNLOAD PDF` or `DOWNLOAD JSON`.

YOU GET The case file, with every input recorded by SHA-256 and both config files by content hash, so the run can be reproduced. Any field the analysis could not establish prints `NOT AVAILABLE` rather than sitting blank. Further down, an INCOIS OOSA / GNOME handoff block gives the release specification a trajectory model needs to start.

03 Getting started

Use the deployed console

Open <https://avanta.spacesdrive.cc>. Nothing to install, and no sign-in. The API sits at <https://avantaapi.spacesdrive.cc>, with an OpenAPI browser at `/docs`.

Three demo scenarios run the whole chain from one click. They exist for different reasons, and the console labels each one:

SCENARIO	MODE	WHAT IT IS FOR
MSC ELSA 3	<code>CACHED</code>	A real Sentinel-1 scene over a real sinking off Kerala, where the source position, time and vessel identity are all public. Start here: the wind was above the detection ceiling, so this scene shows the system refusing to make a claim. It was a sinking, not a deliberate discharge, and AVANTA did not catch this vessel doing anything illegal.
Live	<code>LIVE</code>	The most recent Sentinel-1 pass over the box, with current AIS traffic. It may legitimately return no slick and no candidates. That is a correct result, not a failure.
Synthetic discharge	<code>SYNTHETIC</code>	The enforcement case end to end. The radar scene, the AIS tracks and the forcing are all real; the slick was simulated along one real vessel's track under one forcing realisation and then attributed under a different one, so the true answer is known exactly. Use this one to see attribution work.

Run it yourself

You need Docker with the Compose plugin. Nothing else — the same file that runs the demo is the on-premise deliverable.

```
cp .env.example .env      # then fill in the keys
docker compose up --build
```

The console is then at `http://localhost:5173` and the API at `http://localhost:8000`. All five API keys are free and none is required: with an empty `.env`, the system runs the bundled scene and the bundled AIS capture end to end and badges every panel `FIXTURE`.

04 Reading the results honestly

Three things on screen tell you how much weight the output can carry. Check them before you trust a number.

The data-mode badges

Every panel carries a badge saying where its data came from. They are not decoration — they change what a result means.

BADGE	MEANING
LIVE	Fetches from the upstream source just now.
CACHED	Served from the content-addressed cache. The same bytes as the live fetch, retrieved by content hash rather than re-requested.
FIXTURE	A bundled capture. Real data, but recorded earlier rather than current.
SYNTHETIC	Generated. The slick was simulated, so the ground truth is exact — and the case does not represent a real event.

The wind gate

Oil only produces radar contrast within a band of wind speeds, roughly 3 to 10 m/s. Below that the sea is already glassy and everything looks dark; above it, wind mixes surface oil into the water column and rebuilds the wave field, so a slick that is present may leave no signature at all.

THE MOST IMPORTANT THING IN THIS GUIDE

A scene marked `GATED` is the system **refusing to make a claim**, not failing. Outside the band, finding no oil tells you almost nothing, and the console says so rather than reporting a clean sea. On a gated scene the `FIND CANDIDATE VESSELS` button is disabled and a sentence beneath it names the reason. The card always shows the measured number that made it gate, so you can check the judgement yourself.

Swath coverage

A satellite pass does not always cover the whole box you asked about. If part of it fell outside the swath, the console shows a **Swath coverage** card marked `PARTIAL` with the percentage actually imaged. Read a negative result on a partially imaged box for what it is: most of the area was never looked at, so finding nothing there says very little.

The H0 "unknown source" row

The unknown-source hypothesis is always present in the ranking, and the posterior sums to 1.0 including it. If H0 wins, the board reads **NO ATTRIBUTION — insufficient evidence** and the console refuses to rank vessels above it. This is the behaviour that matters most: remove the true source from the candidate list and ask again, and the system should not simply accuse the next ship along. Across sixteen negative controls in the validation set, it returned no attribution sixteen times.

05 Keyboard shortcuts

Press `?` anywhere to see this list in the console. Shortcuts are ignored while you are typing in a field.

KEY	ACTION
J / K	Move down and up through the alert queue on Watch
Enter	Open the selected scene
R	Run the highlighted scenario
Space	Play or pause the particle timeline on Attribution
E	Open the evidence breakdown
A	Mark attributed
D	Dismiss
?	Show the command reference
Esc	Close the active drawer or dialogue

06 What it does not do

Stated plainly, so you do not have to find these out during a demonstration.

- **No trained neural segmenter is loaded.** The classical detector is the live path and it works; the benchmark dataset a learned model would need did not arrive. The health endpoint and the interface both say *classical detector (no trained checkpoint loaded)*.
- **Attribution is demonstrated on synthetic ground truth.** The real MSC ELSA 3 scene gates on wind, so the live demo shows detection and refusal on real data, and attribution on the synthetic scenario where the answer is known exactly.
- **Live AIS over Indian waters is sparse.** The free crowd-sourced network measures roughly 21 messages per second over the North Sea against 0.1 over the Indian west coast. The live scenario can legitimately return an empty candidate set. An operational deployment would use satellite AIS or the Coast Guard's own feed; swapping the source is a change of collector, not of method.
- **Calibration figures come from synthetic cases.** They describe the method's internal consistency, not field performance against adjudicated MARPOL prosecutions — no corpus of those exists at a scale that would support the claim. The isotonic correction was fitted, made the numbers worse on a held-out split, and is deliberately not applied. The calibration page says so.
- **Not built at all:** NISAR cross-band look-alike filtering, multi-scene slick tracking, CAP 1.2 alert emission and Bonn Agreement volume estimation. The console shows no panel for any of them rather than an empty one.
- **The OOSA handoff is a format, not a connection.** It emits the release specification GNOME needs in the shape INCOIS expects. There is no live link.

For the full claim-by-claim account of what is real, partial, stubbed or planned — each row named against the test behind it — see `CAPABILITY_MATRIX.md` in the repository. `VERIFY.md` carries the acceptance criteria with an honest status per row.

AVANTA · Team Avanta · Smart India Hackathon problem statement SIH26143 · National Technical Research Organisation, Space Technology.
Sentinel-1 data © Copernicus, processed through the Copernicus Data Space Ecosystem. ERA5 and Copernicus Marine products are Copernicus Climate Change Service and Copernicus Marine Service information. AIS positions from aisstream.io. Vessel identity and AIS-off events from Global Fishing Watch, used under their non-commercial terms. Drift and weathering physics are OpenDrift's OpenOil model with the NOAA ADIOS oil database.